

Tensor Analysis

15. Intrinsic or Absolute Derivative

The intrinsic or absolute derivative of A_p along a curve $x^q = x^q(t)$, denoted by $\frac{\delta A_p}{\delta t}$, is defined as the inner product of the covariant derivative of A_p and $\frac{dx^q}{dt}$, that is $A_{p,q} \frac{dx^q}{dt}$, and is given by

$$\frac{\delta A_p}{\delta t} \equiv \frac{dA_p}{dt} - \left\{ \begin{matrix} r \\ pq \end{matrix} \right\} A_r \frac{dx^q}{dt}$$

Similarly, we define

$$\frac{\delta A^p}{\delta t} = \frac{dA^p}{dt} + \left\{ \begin{matrix} p \\ qr \end{matrix} \right\} A^r \frac{dx^q}{dt}$$

The vectors A_p or A^p are said to move parallelly along a curve if their intrinsic derivatives along the curve are zero, respectively.

Intrinsic derivatives of higher rank tensors are similarly defined.